and equivalently,

$$\mathrm{I_d} - X_{n+1} = \gamma(\eta_{n+1} - \xi_{n+1}) \circ X_{n+1}. \quad p_n\text{-a.e.} \tag{A.10}$$

Denote by T_n^π the unique OT map from p_n to π. Expanding $\|X_{n+1} - T_n^\pi\|_{p_n}^2$ as

$$\begin{aligned}
&\|X_{n+1} - T_n^\pi\|_{p_n}^2 \\
&= \|(\mathrm{I_d} - X_{n+1}) - (\mathrm{I_d} - T_n^\pi)\|_{p_n}^2 \\
&= \|\mathrm{I_d} - T_n^\pi\|_{p_n}^2 - 2\langle \mathrm{I_d} - T_n^\pi, \mathrm{I_d} - X_{n+1}\rangle_{p_n} \\
&\quad + \|\mathrm{I_d} - X_{n+1}\|_{p_n}^2 \\
&= \|\mathrm{I_d} - T_n^\pi\|_{p_n}^2 - 2\langle X_{n+1} - T_n^\pi, \mathrm{I_d} - X_{n+1}\rangle_{p_n} \\
&\quad - \|\mathrm{I_d} - X_{n+1}\|_{p_n}^2 \\
&\le \|\mathrm{I_d} - T_n^\pi\|_{p_n}^2 - 2\langle X_{n+1} - T_n^\pi, \mathrm{I_d} - X_{n+1}\rangle_{p_n},
\end{aligned}$$

where in the last inequality we use that $\|\mathrm{I_d} - X_{n+1}\|_{p_n}^2 \ge 0$. By that

$$\|\mathrm{I_d} - T_n^\pi\|_{p_n}^2 = \mathcal{W}_2(p_n, \pi)^2,$$

and together with (A.10), we have

$$\begin{aligned}
&\|X_{n+1} - T_n^\pi\|_{p_n}^2 \le \mathcal{W}_2(p_n, \pi)^2 \\
&\quad - 2\gamma\langle X_{n+1} - T_n^\pi, (\eta_{n+1} - \xi_{n+1}) \circ X_{n+1}\rangle_{p_n}.
\end{aligned} \tag{A.11}$$

Applying Lemma 4.1 with $p = p_n$ and $\rho = p_{n+1}$, we have

$$\begin{aligned}
&G(\pi) - G(p_{n+1}) \\
&\ge \langle T_n^\pi - X_{n+1}, \eta_{n+1} \circ X_{n+1}\rangle_{p_n} + \frac{\lambda}{2}\mathcal{W}_2(p_{n+1}, \pi)^2.
\end{aligned} \tag{A.12}$$

Meanwhile, by Cauchy Schwartz,

$$\begin{aligned}
&|\langle X_{n+1} - T_n^\pi, \xi_{n+1} \circ X_{n+1}\rangle_{p_n}| \\
&\le \|X_{n+1} - T_n^\pi\|_{p_n} \|\xi_{n+1} \circ X_{n+1}\|_{p_n} \\
&\le \varepsilon \|X_{n+1} - T_n^\pi\|_{p_n}
\end{aligned}$$

where the 2nd inequality is by that $\|\xi_{n+1} \circ X_{n+1}\|_{p_n} = \|\xi_{n+1}\|_{p_{n+1}} \le \varepsilon$ (Assumption 1). Since $\lambda > 0$, we have

$$\varepsilon \|X_{n+1} - T_n^\pi\|_{p_n} \le \frac{\varepsilon^2}{\lambda} + \frac{\lambda}{4}\|X_{n+1} - T_n^\pi\|_{p_n}^2. \tag{A.13}$$

Putting together, this gives

$$|\langle X_{n+1} - T_n^\pi, \xi_{n+1} \circ X_{n+1}\rangle_{p_n}| \le \frac{\varepsilon^2}{\lambda} + \frac{\lambda}{4}\|X_{n+1} - T_n^\pi\|_{p_n}^2. \tag{A.14}$$

Inserting (A.12)(A.14) into (A.11) gives

$$\begin{aligned}
&(1 - \frac{\gamma\lambda}{2})\|X_{n+1} - T_n^\pi\|_{p_n}^2 \\
&\le \mathcal{W}_2(p_n, \pi)^2 + 2\gamma\left(G(\pi) - G(p_{n+1}) - \frac{\lambda}{2}\mathcal{W}_2(p_{n+1}, \pi)^2\right) \\
&\quad + \frac{2\gamma}{\lambda}\varepsilon^2.
\end{aligned} \tag{A.15}$$

Because $(X_{n+1}, T_n^\pi)_\# p_n$ is a coupling between p_{n+1} and π, we have

$$\mathcal{W}_2(p_{n+1}, \pi)^2 \le \|X_{n+1} - T_n^\pi\|_{p_n}^2 . \tag{A.16}$$

Under the condition of the lemma, $0 < \gamma\lambda < 2$, and thus $1 - \frac{\gamma\lambda}{2} > 0$ and then the l.h.s. of (A.15) $\ge (1 - \frac{\gamma\lambda}{2})\mathcal{W}_2(p_{n+1}, \pi)^2$. This proves (34). □

Proof of Theorem 4.3. Taking $\pi = q$ and apply Lemma 4.2, by that $2\gamma\,(G(p_{n+1}) - G(\pi)) \ge 0$, (34) gives that for all n,

$$\left(1 + \frac{\gamma\lambda}{2}\right) \mathcal{W}_2^2(p_{n+1}, q) \le \mathcal{W}_2^2(p_n, q) + \frac{2\gamma}{\lambda}\varepsilon^2. \tag{A.17}$$

Define the numbers ρ and α as

$$\rho := \left(1 + \frac{\gamma\lambda}{2}\right)^{-1}, \quad 0 < \rho < 1, \quad \alpha := \sqrt{\frac{2\gamma}{\lambda}}\varepsilon,$$

and define

$$E_n := \mathcal{W}_2(p_n, q)^2,$$

then (A.17) can be written as

$$E_{n+1} \le \rho(E_n + \alpha).$$

Recursively applying from 0 to $n - 1$ gives that

$$E_n \le \rho^n E_0 + \alpha\frac{\rho(1-\rho^n)}{1-\rho} \le \rho^n E_0 + \alpha\frac{\rho}{1-\rho},$$

which by definition is equivalent to (35).

By (35), one will have $\mathcal{W}_2(p_n, q)^2 \le 5\varepsilon^2/\lambda^2$ if

$$\left(1 + \frac{\gamma\lambda}{2}\right)^{-n} \mathcal{W}_2^2(p_0, q) \le \frac{\varepsilon^2}{\lambda^2},$$

which is fulfilled as long as

$$n \ge \frac{2\left(\log \mathcal{W}_2(p_0, q) + \log(\lambda/\varepsilon)\right)}{\log(1 + \gamma\lambda/2)}.$$

This requirement of n is satisfied under (36) by that $0 < \gamma\lambda < 2$ and the elementary relation that $\log(1 + x) \ge x/2$ for $x \in (0, 1)$. We have proved the $\mathcal{W}_2$-error bound.

To show the smallness of the objective gap $G(p_n) - G(q)$, we use (34) again, and by that $\mathcal{W}_2^2(p_{n+1}, q) \ge 0$,

$$2\gamma\,(G(p_{n+1}) - G(q)) \le \mathcal{W}_2^2(p_n, q) + \frac{2\gamma}{\lambda}\varepsilon^2. \tag{A.18}$$

When n already makes $\mathcal{W}_2(p_n, q)^2 \le 5\varepsilon^2/\lambda^2$, we have

$$2\gamma\,(G(p_{n+1}) - G(q)) \le (5 + 2\gamma\lambda)\frac{\varepsilon^2}{\lambda^2} \le 9\frac{\varepsilon^2}{\lambda^2}, \tag{A.19}$$

where in the 2nd inequality we use that $\gamma\lambda < 2$ because $0 < \lambda \le 1$ and $0 < \gamma < 2$. This proves the bound of $G(p_n) - G(q)$ in (37). □

C Proofs in Section 5

C.1 Proofs in Section 5.1.1

Proof of Lemma 5.1. Let $X_1 \sim p$, $X_2 \sim q$, and

$$Y_1 = T(X_1), \quad Y_2 = T(X_2).$$

Then Y_1 and Y_2 also have densities, $Y_1 \sim \tilde{p} := T_\# p$ and $Y_2 \sim \tilde{q} := T_\# q$. By the data processing inequality concerning two probability distributions through the same stochastic transformation for the KL divergence (see, e.g., the introduction of [61]),

$$\mathrm{KL}(\tilde{p}||\tilde{q}) \leq \mathrm{KL}(p||q).$$

In the other direction, $X_i = T^{-1}(Y_i)$, $i = 1, 2$, then data processing inequality also implies

$$\mathrm{KL}(p||q) \leq \mathrm{KL}(\tilde{p}||\tilde{q}).$$

□

Proof of Corollary 5.2. Under Assumption 4, the KL divergence $G(\rho)$ satisfies Assumption 3 (Lemma B.3), also $q \in \mathcal{P}_2^r$ and $G(q) = 0$ is the global minimum of G. The needed assumptions of Theorem 4.3 are all satisfied, by which we have that for the N defined in the corollary,

$$\mathrm{KL}(p_N||q) = G(p_N) \leq \frac{9}{2\gamma}\left(\frac{\varepsilon}{\lambda}\right)^2. \tag{A.20}$$

Under Assumption 1, T_n are all invertible, and thus T_1^N as defined in (38) is invertible. In addition, by Definition 3.1, one can verify that if T_1 and T_2 are non-degenerate, then so is $T_2 \circ T_1$. Using the arguments for $N-1$ times, we have that T_1^N is non-degenerate. Similarly, since each T_n^{-1} is non-degenerate, we have that $(T_1^N)^{-1}$ is also non-degenerate. Now, $p_0 = p$ has density, and then $p_N = (T_1^N)_\# p_0$ also has density (Lemma 3.2). Meanwhile, $q_N = q$ has density (Assumption 4), then $q_0 = ((T_1^N)^{-1})_\# q_N$ also has density. We now have that p_0, q_0, $p_N = (T_1^N)_\# p_0$ and $q_N = (T_1^N)_\# q_0$ all have densities. Then Lemma 5.1 gives that

$$\mathrm{KL}(p_0||q_0) = \mathrm{KL}(p_N||q_N) = \mathrm{KL}(p_N||q),$$

which, by (A.20), is bounded as stated in the corollary. The TV bound is followed by Pinsker's inequality. □

C.2 Proofs in Section 5.1.2

Lemma C.1 (ρ_δ and $\mathcal{W}_2$ closeness)**.** *Suppose $P \in \mathcal{P}_2$, and ρ_t is the density of X_t in an OU process as in* (6)*, then*

(i) $\rho_t \in \mathcal{P}_2^r$ for any $t > 0$,

(ii) $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $\mathcal{W}_2(\rho_\delta, P) < \varepsilon$. In this case, one can choose $\delta \sim \varepsilon^2$.

Proof of Lemma C.1. For the OU process, we have $V(x) = \|x\|^2/2$ in (7). Then for any $t > 0$, $\rho_t = \mathcal{L}_t(P)$ has the expression as, with $\sigma_t^2 := 1 - e^{-2t}$,

$$\rho_t(x) = \int_{\mathbb{R}^d} \frac{1}{(2\pi\sigma_t^2)^{d/2}} e^{-\|x-e^{-t}y\|^2/(2\sigma_t^2)} dP(y). \tag{A.21}$$